$$A_{i \times j} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \cdots & a_{1j} \\ a_{21} & \ddots & \ddots & \vdots \\ a_{31} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \vdots \\ a_{i1} & \cdots & \cdots & a_{ij} \end{bmatrix}$$